

MassEnviroScreen Documentation

UPDATED: July 7, 2025

Introduction

This document presents a tool for identifying or prioritizing the most environmentally vulnerable or burdened communities in Massachusetts based on a cumulative impact score that incorporates exposure to pollution and climate risks and the presence of sensitive or vulnerable populations, as well as whether the community meets criteria related to limited English proficiency, household income, or presence of tribal land. It is intended to act as a supplement to the Massachusetts Environmental Justice Population definition.

This tool - MassEnviroScreen - is a hybrid approach that utilizes elements of the current [Massachusetts Environmental Justice Population definition](#), the presence of [federally recognized tribal lands](#), and a cumulative impact mapping approach. The cumulative impacts mapping approach is modeled on those used by similar tools across the country, especially [California EPA's CalEnviroScreen tool](#) and the [Colorado EnviroScreen tool](#). These tools utilize a 'cumulative impact score' to describe the relative environmental burden of communities across the state and to prioritize those that are most burdened. California defines cumulative impacts as "the exposures, public health or environmental effects from the combined emissions and discharges, in a geographic area, including environmental pollution from all sources, whether single or multi-media, routinely, accidentally, or otherwise released." Impacts consider "sensitive populations and socio-economic factors, where applicable and to the extent data are available." The Colorado tool augments the CalEnviroScreen approach by adding climate risks, which is the approach followed here.

A 'cumulative impact score' is a numerical value that ranks every community (i.e., census block group) on a scale from 0 to 100. Higher values indicate greater cumulative burden. These values also represent percentile ranks, which means that a community's score also indicates the percentage of scores in a group that are equal to or lower than a given score. For example, a census block group with a score of 50 (50th percentile) means that its cumulative impact score is equal to or higher than 50% of census block groups in the state. In this model, we follow California's example of using a score of 75 (the 75th percentile) as one of the thresholds for identifying the most impacted or 'disadvantaged communities.' Census block groups with a minimum score of 75 experience cumulative impacts that are

equal to or higher than 75% of block groups in the state. In other words, these block groups represent the top 25% of cumulative impact scores in Massachusetts.

To define “disadvantaged communities” in this tool, cumulative impact scores of 75 or higher are combined with two of the [existing Massachusetts EJ criteria](#) - limited English speaking households and low income - as well as the presence of federally-recognized tribal territory.

Disadvantaged communities are communities (i.e., census block groups) that meet one or more of the following criteria:

- cumulative impact score (i.e., MassEnviroScore) of 75 or greater, OR
- annual median household income is 65 percent or less of the statewide annual median household income, OR
- 25 percent or more of households identify as speaking English less than “very well”, OR
- occupy or intersect land within the boundaries of Federally Recognized Tribes

MassEnviroScreen scoring methodology

The MassEnviroScreen cumulative impact score model is based on three components representing Pollution and Climate Burden – Exposures and Environmental Effects and Climate Risks – and two components representing Population Characteristics – Sensitive Populations (e.g., in terms of health status and age) and Socioeconomic Factors.

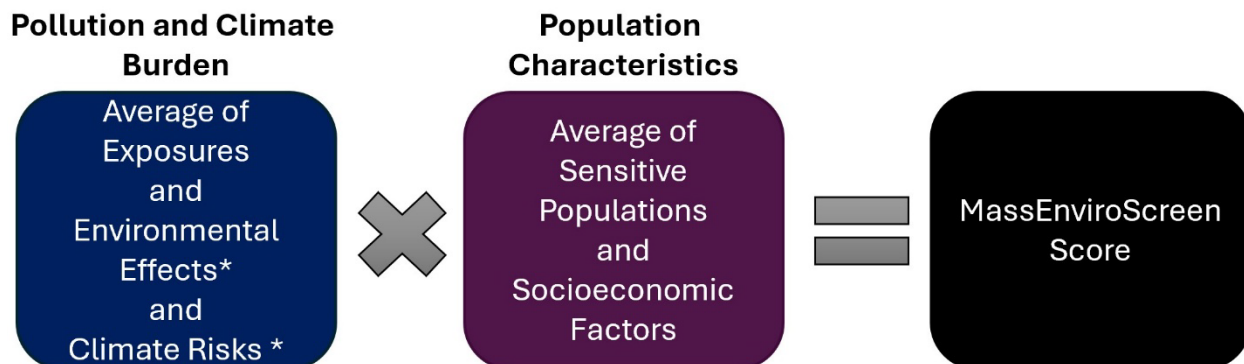
Model characteristics

The model:

- Uses 28 statewide indicators to characterize both Pollution and Climate Burden and Population Characteristics
- Uses percentiles to assign scores for each of the indicators in a given geographic area. The percentile represents a relative score for the indicators.
- Uses a scoring system in which the percentiles are averaged for the set of indicators in each of the four components (Exposures, Environmental Effects, Climate Risks, Sensitive Populations, and Socioeconomic Factors).
- Combines the component scores to produce a MassEnviroScreen score for a given place relative to other places in the state, using the formula below.

Formula for calculating MassEnviroScreen Score

After the components are scored within Pollution Burden and Climate Risk or Population Characteristics, the scores are combined as follows to calculate the overall MassEnviroScreen Score:



* The Environmental Effects and Climate Risks scores were weighted half as much as the Exposures score.

Scores for the pollution burden and climate risk and population characteristics categories are multiplied (rather than added, for example). This multiplication approach is based on research which shows that socioeconomic and health/sensitivity factors are “effect modifiers” that multiply the risks posed by pollutants and climate risks. Various environmental health and emergency response organizations use a similar scoring system: *Risk = Threat x Vulnerability*.

Within the MassEnviroScreen score map, clicking on individual census block groups shows both the final MassEnviroScreen score as well as the Pollution and Climate Burden and the Sensitive Pop SubScores. The Pollution and Climate Burden Subscore represents the percentile rank (0 - 100) of the block group for averaged pollution burden and climate risk component indicators. Similarly, the Sensitive Pop SubScore represents the percentile rank (0 - 100) of the block group for averaged population characteristics component indicators.

Model indicators

The MassEnviroScreen score model is computed from 28 statewide environmental, socioeconomic, and health indicators. These indicators are listed in Table 1 below. Details and data sources for model indicators are described later in this document.

Table 1: MassEnviroScreen Score Indicators

Pollution and Climate Burden	Population Characteristics
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Environmental Exposures	Environmental Effects	Climate Risks	Sensitive Populations	Socioeconomic Factors
PM2.5	Pollution Cleanup Sites	Drought	Pediatric Asthma	Adults without a High School degree
Ozone	Groundwater Threats	Wildfire Risk	Adult High Blood Pressure	Housing Burdened Low Income Households
Nitrogen dioxide (NO2)	Hazardous Waste Generators and Facilities	Flood Risk	Adult Cancer	Linguistic Isolation
Diesel Particulate Matter	Solid Waste Sites and Facilities	Extreme Heat Days	Low Birth Weight Infants	Poverty
Drinking Water Non-compliance	Impaired Water Bodies		Low Life Expectancy	Unemployment
Children's Lead Risk from Housing				
Toxic Releases from Facilities				
Proximity to Heavy Traffic				

MassEnviroScreen Indicator Descriptions

For each indicator, this section provides the following details: Indicator name, definition, short description, which MassEnviroScreen component group it contributes to, the data source, the rationale (why it was selected as an indicator), and method (outlining which assumptions have been made within data processing). For more information about the original datasets, refer to the data source website/s included in each description.

Pollution and Climate Burden - Environmental exposures

PM2.5

Definition: 24-hour average particulate matter that is less than or equal to 2.5 micrometers in diameter (PM2.5) concentrations in parts per million (ppm) during 2021 at census tract centroids. All census block groups received the same PM2.5 measure as the census tract in which they are contained.

Data years: 2021

Originating geographic scale of data: census tracts

Indicator type: Environmental Exposures

Source: U.S. Environmental Protection Agency (EPA) Bayesian Space-time Downscaling Fusion Model (Downscaler) via EJScreen 2024. <https://www.epa.gov/hesc/rsig-related-downloadable-data-files#output>

Ozone (O₃)

Definition: Maximum 8-hour average model predictions in parts per billion (ppb) over the U.S. for ozone for any month in 2021 at census tract centroids. All census block groups received the same Ozone measure as the census tract in which they are contained.

Data years: 2021

Originating geographic scale of data: census tracts

Indicator type: Environmental Exposures

Source: U.S. Environmental Protection Agency (EPA) Bayesian Space-time Downscaling Fusion Model (Downscaler) via EJScreen 2024. <https://www.epa.gov/hesc/rsig-related-downloadable-data-files#output> or <https://screening-tools.com/epa-ejscreen>

Nitrogen dioxide (NO₂)

Definition: average annual nitrogen dioxide (NO₂) levels expressed as part per billion (by volume) for 2020 at 1km grid resolution, aggregated to census block groups using mean pixel values.

Data years: 2020

Originating geographic scale of data: 1km pixels

Indicator type: Environmental Exposures

Source: NASA's Health and Air Quality Applied Sciences Team (HAQAST) - Nitrogen Dioxide Surface-Level Annual Average Concentrations V1 via EJScreen 2024.

https://disc.gsfc.nasa.gov/datasets/SFC_NITROGEN_DIOXIDE_CONC_1/summary?keywords=TROPOMI%20NO2 or <https://screening-tools.com/epa-ejscreen>

Diesel Particulate Matter

Diesel particulate matter (PM) level in air, in micrograms per cubic meter (µg/m³) for 2020 at census tract centroid. All census block groups received the same Diesel PM measure as the census tract in which they are contained.

Data years: 2020

Originating geographic scale of data: census tracts

Indicator type: Environmental Exposures

Source: U.S. Environmental Protection Agency (EPA) Air Toxics Screening Assessment via EJScreen 2024. <https://www.epa.gov/AirToxScreen> or <https://screening-tools.com/epa-ejscreen>

Drinking water Non-Compliance

Safe Drinking Water Act (SDWA) compliance performance of a community water system (CWS) serving a census block group population. The drinking water non-compliance indicator is only applicable to households that get their drinking water from a CWS, which are subject to National Primary Drinking Water Regulations (NPDWRs) under the SDWA. The indicator excludes households on private wells, for example, which are not regulated by the EPA. The indicator reflects compliance performance for the last 5-years (2019 Q4-2023 Q4). Scores are only given to water systems for violations that have not been returned to compliance.

Data years: 2019 - 2023

Originating geographic scale of data: census block groups

Indicator type: Environmental Exposures

Source: EJScreen 2024 at <https://screening-tools.com/epa-ejscreen>

Children's Lead Risk from Housing

Potential risk for lead exposure in children living in low-income communities with older housing. Percentage of households within a census tract with likelihood of lead-based paint (LBP) hazards from the age of housing (1979 or earlier) combined with the percentage of households that are both low-income (household income less than 80% of the county median family income) and have children under 6 years old (5-year estimates 2017-2021). All census block groups received the same percentage as the census tract in which they are contained.

Data years: 2017 - 2021

Originating geographic scale of data: census tracts

Indicator type: Environmental Exposures

Source: U.S. Department of Housing and Urban Development Comprehensive Housing Affordability Strategy data. <https://www.huduser.gov/portal/datasets/cp.html>

Toxic Releases from Facilities

Definition: Average annual Risk-Screening Environmental Indicators (RSEI) modeled toxicity-weighted concentrations in air of Toxics Release Inventory (TRI) listed chemicals to census block groups. Quantifies relative potential human health impacts of certain chemicals included on the list of toxic chemicals established under section 313 of the Emergency Planning and Community Right-to-Know Act (EPCRA) based on the quantities released to air by facilities. The indicator is derived by considering the quantities of the chemical(s) released to air, modeling the fate of the chemical(s) in air based on an approximate point of release, determining chemical concentrations in air, and weighting the chemical concentrations by the relative toxicity(ities) of the chemical(s). Average annual toxicity-weighted concentration results from all modeled air releases are aggregated at the U.S. Census block group.

Data years: 2020 population data, 2022 TRI reporting

Originating geographic scale of data: census block groups

Indicator type: Environmental Exposures

Source: EPA Risk-Screening Environmental Indicators (RSEI) model Geographic Microdata via EJScreen 2024. <https://www.epa.gov/rsei/rsei-geographic-microdata-rsei-gm> or <https://screening-tools.com/epa-ejscreen>

Proximity to Heavy Traffic

Definition: Count of vehicles (annual average daily traffic [AADT]) at major roads within 10 kilometers (km), divided by distance in km. The traffic proximity indicator is based on AADT count divided by distance in meters from the Census block centroid. The proximity score is based on the traffic within a search radius of 10 km. The closest traffic is given more weight, and the distant traffic is given less weight, through inverse distance weighting. The traffic proximity indicator values for each block centroid are averaged to the census block group in which they are contained.

Data years: 2020 HPMS release, Federal Highway Administration (FHWA), U.S. Department of Transportation (DOT)

Originating geographic scale of data: census block groups

Indicator type: Environmental Exposures

Source: EJScreen 2024 at <https://screening-tools.com/epa-ejscreen>

Pollution and Climate Burden - Environmental effects

Pollution Cleanup Sites

Definition: Sum of weighted sites within each census block group, adapted from methodology used for CalEnviroScreen. Sites were scored on a weighted scale of 0 to 12 in consideration of both the site type and status. Higher weights were applied to Superfund, state response sites, and cleanups compared to evaluations, for example. Similarly, higher weights were applied to sites that are undergoing active remediation and oversight, relative to those with little or no state involvement. The weights for all sites were adjusted based on the distance they fell from populated census blocks. Sites further than 1000m from any populated census block were excluded from the analysis. Site weights were adjusted by multiplying the weight by 1 for sites less than 250m, 0.5 for sites 250-500m, 0.25 for sites 500-750m, and 0.1 for sites 750-1000m from the nearest populated census blocks within a given census block group. Each census block group was scored based on the sum of the adjusted weights. Summed census block group scores were sorted and assigned percentiles based on their position in the distribution. The table below shows the weights for each site type.

Data years: National Priorities List and Superfund Alternative Approach Sites 2024; EPA ACRES brownfield sites 2024; MA DEP 21E sites 2025; MA DEP AUL sites 2025

Originating geographic scale of data: point locations

Indicator type: Environmental Effects

Weighting Matrix for Cleanup Sites

Site Type	Status		
	Low • Certified • Completed • No Further Action	Medium • Inactive – Needs Eval. • Inactive • Certified Operation & Maintenance – Land Use Restrictions • Certified Operation & Maintenance	High • Active • Backlog • Inactive – Action Required
Low • Evaluation • Historical • Military Eval	0	4	6
Medium • Corrective Action • School Cleanup • Voluntary Cleanup • Tiered Permit	1	7	9
High • State Response • Superfund	2	10	12

Active: Identifies that an investigation and/or remediation is currently in progress and that DEP is actively involved, either in a lead or support capacity.

Certified Operation and Maintenance (O&M): Identifies sites that have certified cleanups in place but require ongoing O&M activities.

Certified: Identifies completed sites with previously confirmed releases that are subsequently certified by DEP as having been remediated satisfactorily under DEP oversight.

Corrective Action: Identifies sites undergoing “corrective action,” defined as investigation and cleanup activities at hazardous waste facilities (either Resource Conservation and

Recovery Act (RCRA) or State-only) that either were eligible for a permit or received a permit. These facilities treat, store, dispose and/or transfer hazardous waste.

Evaluation: Identifies suspected, but unconfirmed, contaminated sites that need or have gone through a limited investigation and assessment process. Includes brownfield sites.

Inactive – Action Required: Identifies non-active sites where DEP has determined that a removal or remedial action or further extensive investigation is required.

Inactive - Needs Evaluation: Identifies inactive sites where DEP has determined an evaluation is required.

No Further Action: Identifies completed sites where DTSC determined after investigation, generally a PEA (an initial assessment), that the property does not pose a problem to public health or the environment.

School Cleanup: Identifies proposed and existing school sites that are being evaluated by DEP for possible hazardous materials contamination at which remedial action occurred.

State Response: Identifies confirmed release sites where DEP is involved in remediation, either in a lead or oversight capacity. These confirmed release sites are generally high-priority and high potential risk. This include 21E sites.

Superfund: Identifies sites where the US EPA proposed, listed, or delisted a site on the National Priorities List (NPL).

Tiered Permit Sites: These facilities manage waste not regulated under RCRA, but regulated as a hazardous waste by the State of Massachusetts.

Voluntary Cleanup: Identifies sites with either confirmed or unconfirmed releases, and the project proponents have requested that DEP oversee evaluation, investigation, and/or cleanup activities and have agreed to provide coverage for DEP's costs.

Sources: Superfund National Priorities List polygons from US EPA Office of Land and Emergency Management (OLEM) at https://edg.epa.gov/data/PUBLIC/OLEM/OLEM-OSRTI/NPL_Boundaries.zip; Brownfields sites from US EPA Assessment, Cleanup and Redevelopment Exchange System (ACRES) at <https://catalog.data.gov/dataset/acres-brownfields-properties>; MA DEP 21E sites from MassGIS at <https://www.mass.gov/info-details/massgis-data-massdep-tier-classified-oil-andor-hazardous-material-sites-mgl-c-21e>; MA DEP AUL sites from MassGIS at <https://www.mass.gov/info-details/massgis-data-massdep-oil-andor-hazardous-material-sites-with-activity-and-use-limitations-aul>;

Groundwater Threats

Sum of weighted scores for sites within each census block group, adapted from methodology used for CalEnviroScreen. Sites include USTs and facilities with groundwater discharge permits. The weights for all sites were adjusted based on their distance from populated census blocks. Sites further than 1000m from any populated census block were excluded from the analysis. Site weights were adjusted by multiplying the weight by 1 for sites less than 250m, 0.5 for sites 250-500m, 0.25 for sites 500-750m, and 0.1 for sites 750-1000m from the nearest populated census blocks within a given block group. Sites outside of a census block group, but less than 1000m from one of that block group's populated blocks were similarly adjusted based on the distance to the nearest block from that block group. Each census block group was scored based on the sum of the adjusted weights for sites it contains or is near.

Data years: US EPA UST Finder 2018-2021; MA DEP Groundwater Discharge Permits 2025

Originating geographic scale of data: point locations

Indicator type: Environmental Effects

Sources: US EPA's UST Finder data is a national composite of leaking underground storage tanks, underground storage tank facilities, and underground storage tanks as of 2018-2021.

Data downloaded via ArcGIS Pro at

<https://epa.maps.arcgis.com/home/item.html?id=5a3ae0ed53564b6fa519f08e30e79e93>;

MA DEP Groundwater Discharge Permits from MassGIS at <https://www.mass.gov/info-details/massgis-data-massdep-groundwater-discharge-permits>.

Weighting Matrix for Groundwater Threats

Site Type	Status	Weight
Groundwater Discharge Permit Facilities	Active	5
	Inactive	2
UST Sites	Open	3
	No Further Action	3
	Unknown	3

Hazardous Waste Generators and Facilities

Definition: Sum of weighted permitted hazardous waste facilities, and hazardous waste generators within each census block group. The weights for all facilities were adjusted based on the distance they fell from populated census blocks. All facilities further than 1,000m from any populated census block were excluded from the analysis. Site weights were adjusted by multiplying the weight by 1 for facilities less than 250m, 0.5 for sites 250-500m, 0.25 for sites 500-750m, and 0.1 for sites 750-1000m from the nearest populated

census blocks within a given tract. Facilities outside of a census tract, but less than 1000m from one of that tract's populated blocks were similarly adjusted based on the distance to the nearest block from that tract. Each census tract was scored based on the sum of the adjusted weights for sites it contains or is near.

Data years: 2024

Originating geographic scale of data: point locations

Indicator type: Environmental Effects

Source: MassDEP Major Facilities from MassGIS at <https://www.mass.gov/info-details/massgis-data-massdep-major-facilities>.

Weighting Matrix for Permitted Hazardous Waste Facilities, Hazardous Waste Generators, and Chrome Plating Facilities

Permitted Hazardous Waste Facilities

	Weight	Activity or Status
Facility Activity (base weight)	10	Landfill
	7	Treatment
	4	Storage
	2	Post-closure
Permit Type (additional weight)	1	Large facilities
	1	Non-RCRA facilities
	2	RCRA facilities

Hazardous Waste Generators

Generator Type	Weight	Quantity of Waste
Large Quantity Hazardous Waste Generators	0.1	< 100 tons/yr
	0.5	100 – 1,000 tons/yr
	2	>1,000 tons/yr

Solid Waste Sites and Facilities

Definition: Sum of weighted solid waste sites and facilities. The weights for all sites, including the large landfill perimeters, were adjusted based on the distance they fell from populated census blocks. Sites further than 1000m from any populated census block were excluded from the analysis. Site weights were adjusted by multiplying the weight by 1 for sites less than 250m, 0.5 for sites 250-500m, 0.25 for sites 500-750m, and 0.1 for sites 750-1000m from the nearest populated census blocks within a given census block group. Sites

outside of a census block group, but less than 1000m from one of that tract's populated blocks were similarly adjusted based on the distance to the nearest populated block from that block group. Each census block group was scored based on the sum of the adjusted weights for sites it contains or is near. Summed census block group scores were sorted and assigned percentiles based on their position in the distribution.

Data years: 2025

Originating geographic scale of data: point locations

Indicator type: Environmental Effects

Source: MassDEP Solid Waste Diversion and Disposal layer from MassGIS at <https://www.mass.gov/info-details/massgis-data-massdep-solid-waste-diversion-and-disposal>.

Weighting Matrix for Solid Waste Sites and Facilities

Category	Site of Facility Type
Unpermitted Dumping Grounds (inactive)	6
Solid Waste Landfill or Construction, Demolition and Inert (CDI) Debris Waste Disposal (active)	8
Solid Waste Disposal Site, historic incinerator or combustion facility (closed, inactive)	1
Combustion facility (active)	10
Composting	4 Site assigned composting facility 2 Site assignment exempt composting facility
Aerobic/Anaerobic Digesting	3
Transfer, processing, handling facility	5 Permitted large volume, C&D 2 Permitted small volume, organic or wood waste
Waste Tire, recycling, wood waste	4

Impaired Water Bodies

Definition: Summed number of pollutants across all water bodies designated as impaired within the area, adapted from methodology used for CalEnviroScreen. The number of pollutants listed in streams or rivers that fell within 1 kilometer (km) or 2 km respectively of a census block group's populated blocks were counted. The 2 km buffer distance was applied to major rivers (>100 km in length). The 1 km buffer distance was applied to all

smaller streams/rivers. The number of pollutants listed in lakes, bays, estuaries or shoreline that fell within 1 km or 2 km of a census block group's populated blocks were counted. The 2 km buffer distance was applied to major lakes or bays greater than 25 square kilometers in size. The 1 km buffer distance was applied for all other lakes/bays. The two pollutant counts were summed for every census block group. Each census block group was scored based on the sum of the number of individual pollutants found within and/or bordering it. For example, if two stream sections within a census block group were both listed for the same pollutant, the pollutant was only counted once. Summed census block group scores were sorted and assigned percentiles based on their position in the distribution.

Data years: 2022

Originating geographic scale of data: water body linear and polygon features

Indicator type: Environmental Effects

Source: MassDEP 2022 Integrated List of Waters (305(b)/303(d)) from MassGIS at <https://www.mass.gov/info-details/massgis-data-massdep-2022-integrated-list-of-waters-305b303d>.

Pollution and Climate Burden - Climate Risks

Drought

Definition: Sum of weekly total percent of an area experiencing a severe, extreme, or exceptional drought (categories D2, D3, or D4), adapted from Colorado EnviroScreen. These levels of drought imply some level of voluntary or mandated water use restrictions and observable damage or loss of pasture and crops. The U.S. Drought Monitor reports the percentage of each county experiencing each of the six drought levels (None, D0, D1, D2, D3, and D4) each week. The sum of areas experiencing D2, D3, D4 level droughts was calculated weekly across all weeks from January 2019 to December 2024. The sum of the weekly drought values across that time period was used to define the Drought measure. All census block groups received the Drought value for the county in which they are located.

Data years: 2019 - 2024

Originating geographic scale of data: county

Indicator type: Climate Risks

Source: U.S. Drought Monitor 2019-2024 <https://droughtmonitor.unl.edu/Data.aspx>

Wildfire Risk

Definition: Wildfire Hazard Potential for the United States, version 2023. The mean wildfire hazard potential within each geographic area is used as the Wildfire risk score, adapted from Colorado EnviroScreen. Wildfire Hazard Potential (WHP) is provided as a 270-meter resolution raster. The mean WHP of all pixels or raster grid cells falling within each census block group were calculated.

Data years: 2023

Originating geographic scale of data: 270-meter pixels

Indicator type: Climate Risks

Source: U.S. Department of Agriculture (USDA), U.S. Forest Service (USFS)
<https://www.fs.usda.gov/rds/archive/catalog/RDS-2015-0047-4>

Flood Risk

Definition: Percentage of each geographic area where there is at least a one percent chance of flooding annually, adapted from Colorado EnviroScreen. The area of all features within the 1% Annual Chance Flood Hazard within each census block group was divided by the total area of the census block group. If no flood areas were found within the census block group, a value of zero was used. Note that the NFHL digital FIRMs are not available for the northwest quadrant of Massachusetts. These areas were supplemented with Q3 FEMA Q3 Flood Zones from Paper FIRMs data. Franklin County is not available in either of the preceding datasets, and was supplemented with "riverine flooding - exposure - impacted area" from FEMA NRI at census tract level, downscaled to block groups.

Data years: FEMA NFHL data 2023; FEMA Q3 Flood Zones from Paper FIRMs data 1997 - 2013; FEMA National Risk Index data for Franklin County 2025

Originating geographic scale of data: delineated flood polygons for NFHL and Q3 data; census tracts for Franklin County

Indicator type: Climate Risks

Source: FEMA National Flood Hazard Layer (NFHL) from MassGIS at
<https://www.mass.gov/info-details/massgis-data-fema-national-flood-hazard-layer;>

Extreme Heat Days

Definition: Average number of days between May and September from 2019 through 2023 in which daily high temperature exceeded the 90th percentile of historical daily high

temperatures, adapted from Colorado EnviroScreen. The mean number of extreme heat days between 2019-2023 was available at the census tract and county level. All census block groups received the Extreme heat days value for the census tract in which they are located.

Data years: 2019 – 2023

Originating geographic scale of data: census tracts

Indicator type: Climate Risks

Source: National Environmental Public Health Tracking Network via the U.S. Centers for Disease Control (CDC), Heat & Heat Related Illness (HRI), Historical Temperature & Heat Index, 2019-2023 <https://ephtracking.cdc.gov/>

Population Characteristics - Sensitive populations

Pediatric Asthma

Definition: Pediatric asthma prevalence per 100 students (K-8) by school and community for years 2017 to 2023. Values provided by municipality (i.e., city or town). All census block groups received the prevalence value for the municipality in which they are located.

Data years: 2017 – 2023

Originating geographic scale of data: municipality

Indicator type: Sensitive Populations

Source: Massachusetts Environmental Public Health Tracking at <https://matracking.ehs.state.ma.us/Health-Data/Asthma/index.html>

Adult High Blood Pressure

Definition: Prevalence of high blood pressure among adults from 2024. Data provided a census tract level. All census block groups received the prevalence value for the census tract in which they are located.

Data years: 2024

Originating geographic scale of data: census tracts

Indicator type: Sensitive Populations

Source: CDC PLACES Health Outcomes at https://data.cdc.gov/500-Cities-Places/PLACES-Local-Data-for-Better-Health-Census-Tract-D/cwsq-ngmh/about_data

Adult Cancer

Definition: Prevalence of cancer (non-skin) or melanoma among adults from 2024. Data provided a census tract level. All census block groups received the prevalence value for the census tract in which they are located.

Data years: 2024

Originating geographic scale of data: census tracts

Indicator type: Sensitive Populations

Source: CDC PLACES Health Outcomes at https://data.cdc.gov/500-Cities-Places/PLACES-Local-Data-for-Better-Health-Census-Tract-D/cwsq-ngmh/about_data

Low Birth Weight Infants

Definition: Percentage of live births that were low or very low birth weight for 2013 to 2022. Data provided at municipality level (i.e., town or city). All census block groups received the prevalence value for the municipality in which they are located.

Data years: 2013 – 2022

Originating geographic scale of data: municipality

Indicator type: Sensitive Populations

Source: Massachusetts Department of Public Health Birth Outcomes Data of Massachusetts Residents at <https://www.mass.gov/info-details/birth-outcomes-data-of-massachusetts-residents>

Elevated Blood Lead Levels in Children

Definition: Average annual prevalence of males and females with estimated confirmed blood lead levels ≥ 5 $\mu\text{g}/\text{dL}$ (estimated confirmed) in 2017 - 2021 that were between 9 and less than 48 months of age by census tract. All census block groups received the prevalence value for the census tract in which they are located.

Data years: 2017 – 2021

Originating geographic scale of data: census tracts

Indicator type: Sensitive Populations

Source: Massachusetts Environmental Public Health Tracking via MassDEP Cumulative Impact Analysis in Air Quality Permitting at <https://www.mass.gov/info-details/cumulative-impact-analysis-in-air-quality-permitting#cia-guidance-and-tools->

Premature Mortality (PM)

Definition: Age-adjusted premature mortality rate (PMR per 100,000) by census tract. Premature mortality tracks unfulfilled life expectancy and is based on deaths prior to age 75 (premature). PM includes unintentional injuries such as motor vehicle-related deaths, poisonings, falls, fires, and drownings that were not intended to occur. Deaths are for all age groups. Massachusetts Department of Public Health (MDPH) provided an age-adjusted premature mortality rate (PMR per 100,000) by tract. MDPH calculated the statewide PMR (292.5 per 100,000) and MassDEP calculated the percent of state rate and percentile (percentile rank) from the tract values. All census block groups received the mortality rate for the census tract in which they are located.

Data years: 2018 – 2022

Originating geographic scale of data: census tracts

Indicator type: Sensitive Populations

Source: Massachusetts Department of Public Health Death Data Registry of Vital Records and Statistics acquired via MassDEP Cumulative Impact Analysis in Air Quality Permitting at <https://www.mass.gov/info-details/cumulative-impact-analysis-in-air-quality-permitting#cia-guidance-and-tools->

Population Characteristics - Socioeconomic Factors

Adults Without a High School Diploma

Definition: Percent of people age 25 or older in a block group whose education is less than a high school diploma. The ACS education information is captured in the table Sex by Educational Attainment for the Population 25 Years and Over (ACS Table ID: B15002) at the census block group level.

Data years: 2019 – 2023

Originating geographic scale of data: census block groups

Indicator type: Socioeconomic Factors

Source: US American Community Survey 5-year Estimates for 2019 – 2023.

Housing Burdened Low Income Households:

Definition: Percent of households that are both low income (making less than 80% of the HUD Area Median Family Income) and severely burdened by housing costs (paying greater than 50% of their income to housing costs). Data provided at census tract level. All census block groups received the percentage value for the census tract in which they are located.

Data years: 2019 – 2023

Originating geographic scale of data: census block groups

Indicator type: Socioeconomic Factors

Source: US Department of Housing and Urban Development CHAS (Comprehensive Housing Affordability Strategy) at <https://www.huduser.gov/portal/datasets/cp.html>

Linguistic Isolation

Definition: Percentage of limited English-speaking households, (2019-2023). A limited English speaking household is defined as a household in which no one age 14 or over speaks English at least "very well" as reported in the U.S. Census Bureau's ACS. The ACS limited English speaking household information is captured in the table Household Language by Household Limited English Speaking Status (ACS Table ID: C16002) at the census block group level.

Data years: 2019 – 2023

Originating geographic scale of data: census block groups

Indicator type: Socioeconomic Factors

Source: US American Community Survey 5-year Estimates for 2019 – 2023

Poverty

Definition: Percent of household whose income is less than or equal to twice the poverty level (2019 – 2023). For example, a household of four with a reported \$40,000 total annual income is lower than twice the poverty threshold of \$59,900 (\$29,950 is the poverty threshold defined by the U.S. Census Bureau for 2022). The poverty level is a national number and the same across all geographic regions. To accommodate differences in the varying costs of living across the United States and other factors, analysts typically use twice the poverty level to capture low-income households, especially in high-cost areas. The ACS low income information is captured in the table Ratio of Income to Poverty Level in the Past 12 Months (ACS Table ID: C17002).

Data years: 2019 – 2023

Originating geographic scale of data: census block groups

Indicator type: Socioeconomic Factors

Source: US American Community Survey 5-year Estimates for 2019 – 2023.

Unemployment

Definition: Percentage of the population over the age of 16 that is unemployed and eligible for the labor force. Excludes retirees, students, homemakers, institutionalized persons except prisoners, those not looking for work, and military personnel on active duty (5-year estimate, 2019-2023). The ACS unemployment information is captured in the table Employment Status for the Population 16 Years and Over (Table ID: B23025).

Data years: 2019 – 2023

Originating geographic scale of data: census block groups

Indicator type: Socioeconomic Factors

Source: US American Community Survey 5-year Estimates for 2019 – 2023.